Scotch Student ID #			
Circle the relevant digits			
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Teacher's Name

Scotch College

CHEMISTRY

SAC 1 – FUELS, GASES & THERMOCHEMISTRY

Part B

Date: Tuesday, March 10th

Reading Time	0 minutes
Writing Time	45 minutes

Task Sections	Marks	Your Marks
Total Marks	40	

General Instructions
• Write your student number in the space provided above on this page. • Please complete all questions in the spaces provided.
Allowed Materials
• A VCAA Chemistry Data Book has been provided and you may use your Unit 3 SAC 1 logbook from Part A • Students are permitted to bring into the SAC: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.
At the end of the task…
• Please ensure your student number has been filled in at the top of the page and that you have written your teacher's name.
Electronic Devices
Students are **not** allowed to have a mobile phone, smart watch and/or any other unauthorised electronic device in the SAC, unless it is TURNED OFF and is placed on the front teacher desk.

Section A (12 marks)

Q1. Summary of analysis (refer to logbook for working) (6 marks)

formula of alcohol	
mass of fuel burned (g)	
amount, in mol, of fuel burned (mol)	
Energy, q, absorbed by water (J)	
Energy, q, released by combustion of the alcohol (J)	
Energy released per mol of fuel during combustion (molar heat of combustion) (kJ mol^{-1})	

Q2. Based on your results, write a thermochemical equation for the complete combustion of the alcohol. (3 marks)

_____

Q3. Discuss the reliability of your data and suggest **two** ways in which the experimental design might be improved.

(3 marks)

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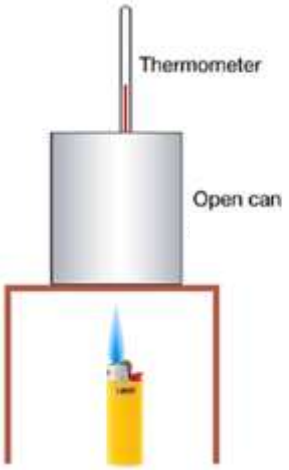
Section B

Q4. *(10 marks)*

Butane, C_4H_{10}, is a liquid hydrocarbon that is used as a fuel. A student designed an experiment similar to yours to measure the heat of combustion of butane, as shown in the diagram.



Mass of water in can = 200 g
Mass of butane burnt = 1.50 g
Initial temperature of water = 18.2°C
Final temperature of water = 63.4°C

a. Use the temperature readings provided to calculate the temperature change (ΔT) in the water in the can. (1 mark)

_____

b. Use the specific heat capacity of water (from the data book) and ΔT to determine the amount of energy that has been transferred to the water. (1 mark)

_____

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c. Write a balanced chemical equation for the complete combustion of butane, C_4H_{10}. (2 marks)

_____

d. Calculate the amount, in mole, of butane that underwent combustion during the experiment. (1 mark)

_____

_____

_____

e. Use your answers to calculate Δ*H* for the combustion reaction of butane according to the equation given in question (c). (3 marks)

_____

_____

_____

f. By comparing your results with the literature molar heats of combustion given below, give two sources of experimental error that can account for the discrepancy between the experimental result obtained here and the value provided in the data book. (2 marks)

Alcohol	**Molar heat of combustion ($kJ\ mol^{-1}$)**
methanol	726
ethanol	1360
propanol	2016

_____

_____

_____

_____

Q5. *(8 marks)*

Butane is a component of liquefied petroleum gas (LPG) which is used to power some cars. A tank of butane can supply 2500 MJ of energy for driving.

a. Use Table 11 in the data book to determine the mass of butane needed to supply this much energy. (2 marks)

_____

_____

_____

b. Use your answer from 5a to determine the mass of carbon dioxide emitted into the atmosphere during the combustion of this mass of butane. (3 marks)

_____

_____

_____

c. In another experiment, 2100 MJ of energy was produced by a tank of butane. This produced 128 kg of CO_2 gas. Determine the mass of CO_2 emitted per MJ of energy released from combustion of butane. (1 mark)

_____

_____

_____

d. Biogas, which is predominantly made of methane, is used as a fuel. The equation for complete combustion of methane is:

$$CH_4\,(g) + 2O_2\,(g) \rightarrow CO_2\,(g) + 2H_2O\,(g)$$

The combustion of enough methane to give 3000 MJ produces 148.3 kg of CO_2. Determine the volume, in litres, that this amount of CO_2 would occupy at standard laboratory conditions (SLC). (2 marks)

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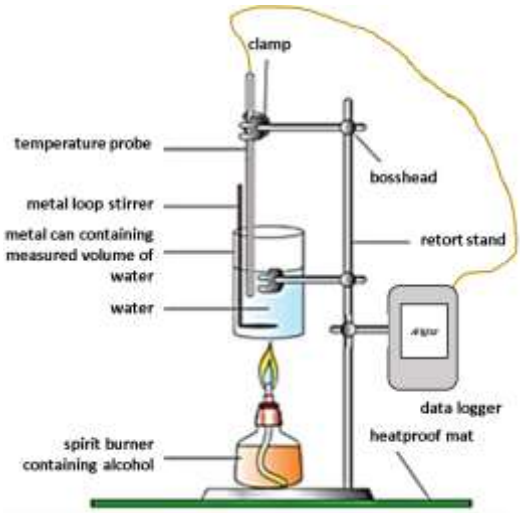
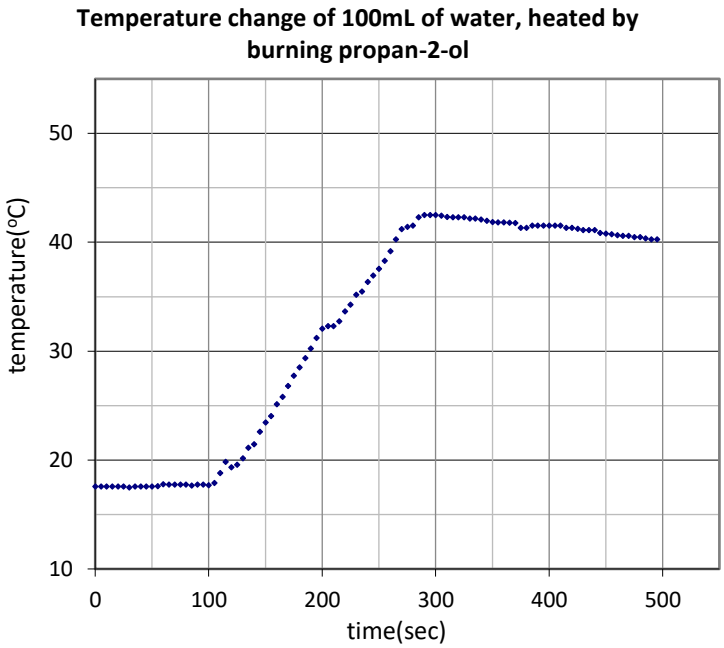
Q6. (4 marks)
Identify the advantages and disadvantages of using natural gas and bioethanol for fuel by comparing the two in terms of their energy content, net effects on atmospheric carbon dioxide levels, renewability and sustainability.

Fuel Type	Energy content (Use databook values)	Net CO_2 levels	Renewability	Sustainability
Bioethanol				
Natural gas				

Q7. *(6 marks)*

Another student used the same apparatus as you have used but substituted the thermometer with a temperature probe for more accurate recordings with the same aim as the experiment you conducted. The burner was lit after 100 seconds and extinguished at 270 seconds. The fuel used was propan-2-ol. Below is their experimental setup and a graph of their data.





Mass of water in can = 100 g
Mass of propan-2-ol burnt = 1.379 g
Initial temperature of water = 17.7°C
Maximum temperature of water = 42.5°C

a. Explain why the temperature of the water increased between 100 and 270 seconds. (1 mark)

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b. Explain why the temperature gradually decreased between 300 and 500 seconds. (1 mark)

_____

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c. Would you expect the result (ΔH, *in kJ mol*$^{-1}$) to be similar or quite different if butan-1-ol was used instead? Briefly explain your reasoning. (2 marks)

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d. Write a concise conclusion for this student's experiment. (2 marks)

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END OF SAC